No 1

```
root@jichi:~# ip -br addr; ip route; cat /proc/sys/net/ipv4/ip_forward; iptables
 -t nat -S POSTROUTING
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0             UNKNOWN        192.168.122.2/24 fe80::42:dfff:fe69:7000/64
eth1             UNKNOWN        192.227.1.1/24 fe80::42:dfff:fe69:7001/64
eth2             UNKNOWN        192.227.2.1/24 fe80::42:dfff:fe69:7002/64
eth3             UNKNOWN        192.227.3.1/24 fe80::42:dfff:fe69:7003/64
eth4             UNKNOWN        192.227.4.1/24 fe80::42:dfff:fe69:7004/64
default via 192.168.122.1 dev eth0
192.168.122.0/24 dev eth0 proto kernel scope link src 192.168.122.2
192.227.1.0/24 dev eth1 proto kernel scope link src 192.227.1.1
192.227.2.0/24 dev eth2 proto kernel scope link src 192.227.2.1
192.227.3.0/24 dev eth3 proto kernel scope link src 192.227.3.1
192.227.4.0/24 dev eth4 proto kernel scope link src 192.227.4.1
1
-P POSTROUTING ACCEPT
-A POSTROUTING -s 192.227.0.0/16 -o eth0 -j MASQUERADE
```

```
bash /root/soal1.shm0 > /etc/resolv.confkenal"; exit 1 ;;P=$P.4.10 ;;
eth0             UNKNOWN        192.227.4.5/24 fe80::42:c5ff:fe41:c800/64
Gateway: 192.227.4.1
PING 192.227.4.1 (192.227.4.1) 56(84) bytes of data.
64 bytes from 192.227.4.1: icmp_seq=1 ttl=64 time=1.17 ms
64 bytes from 192.227.4.1: icmp_seq=2 ttl=64 time=0.506 ms
64 bytes from 192.227.4.1: icmp_seq=3 ttl=64 time=0.555 ms

--- 192.227.4.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2006ms
rtt min/avg/max/mdev = 0.506/0.742/1.166/0.300 ms
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=111 time=26.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=111 time=25.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=111 time=25.6 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 25.535/25.722/26.002/0.201 ms
PING google.com (172.217.194.139) 56(84) bytes of data.
64 bytes from si-in-f139.1e100.net (172.217.194.139): icmp_seq=1 ttl=101 time=25
.2 ms
64 bytes from si-in-f139.1e100.net (172.217.194.139): icmp_seq=2 ttl=101 time=25
.1 ms
64 bytes from si-in-f139.1e100.net (172.217.194.139): icmp_seq=3 ttl=101 time=25
.6 ms

--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 25.094/25.304/25.645/0.243 ms
root@panda:~#
```

No 2

sed -i -E 's/tengen\)(+)IP=\$P\.4\.2 /tengen)\1IP=$P.4.10 /; s/inumaki\)(+)IP=\$P\.4\.10 /inumaki)\1IP=$P.4.2 /' /root/soal1.sh

ip addr flush dev eth0

bash /root/soal1.sh

cp /root/soal1.sh /etc/network/soal1.sh

```
root@tengen:~# ip addr flush dev eth0
root@tengen:~# bash /root/soal1.sh
eth0             UNKNOWN        192.227.4.10/24
Gateway: 192.227.4.1
PING 192.227.4.1 (192.227.4.1) 56(84) bytes of data.
64 bytes from 192.227.4.1: icmp_seq=1 ttl=64 time=0.686 ms
64 bytes from 192.227.4.1: icmp_seq=2 ttl=64 time=0.360 ms
64 bytes from 192.227.4.1: icmp_seq=3 ttl=64 time=0.361 ms

--- 192.227.4.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2272ms
rtt min/avg/max/mdev = 0.360/0.469/0.686/0.153 ms
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=111 time=22.7 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=111 time=24.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=111 time=22.7 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2218ms
rtt min/avg/max/mdev = 22.717/23.305/24.480/0.830 ms
PING google.com (172.217.194.100) 56(84) bytes of data.
64 bytes from si-in-f100.1e100.net (172.217.194.100): icmp_seq=1 ttl=101 time=23
.7 ms
64 bytes from si-in-f100.1e100.net (172.217.194.100): icmp_seq=2 ttl=101 time=23
.6 ms
64 bytes from si-in-f100.1e100.net (172.217.194.100): icmp_seq=3 ttl=101 time=24
.1 ms

--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2218ms
rtt min/avg/max/mdev = 23.637/23.807/24.096/0.205 ms
root@tengen:~# cp /root/soal1.sh /etc/network/soal1.sh
```

```
root@inumaki:~# bash /root/soal1.sh
eth0             UNKNOWN        192.227.4.2/24
Gateway: 192.227.4.1
PING 192.227.4.1 (192.227.4.1) 56(84) bytes of data.
64 bytes from 192.227.4.1: icmp_seq=1 ttl=64 time=0.703 ms
64 bytes from 192.227.4.1: icmp_seq=2 ttl=64 time=0.476 ms

64 bytes from 192.227.4.1: icmp_seq=3 ttl=64 time=0.488 ms

--- 192.227.4.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2259ms
rtt min/avg/max/mdev = 0.476/0.555/0.703/0.104 ms
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=111 time=23.6 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=111 time=23.0 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=111 time=23.0 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2220ms
rtt min/avg/max/mdev = 22.980/23.221/23.636/0.294 ms
PING google.com (172.217.194.100) 56(84) bytes of data.
64 bytes from si-in-f100.1e100.net (172.217.194.100): icmp_seq=1 ttl=101 time=2
.3 ms
64 bytes from si-in-f100.1e100.net (172.217.194.100): icmp_seq=2 ttl=101 time=2
.6 ms
64 bytes from si-in-f100.1e100.net (172.217.194.100): icmp_seq=3 ttl=101 time=2
.1 ms

--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2219ms
rtt min/avg/max/mdev = 23.147/23.692/24.615/0.656 ms
root@inumaki:~# cp /root/soal1.sh /etc/network/soal1.sh
root@inumaki:~#
```

```
  isc-dhcp-common isc-dhcp-server policycoreutils selinux-utils
0 upgraded, 4 newly installed, 0 to remove and 29 not upgraded.
Need to get 1832 kB of archives.
After this operation, 8002 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian trixie/main amd64 isc-dhcp-common amd64 4.4.3
-P1-8 [118 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 isc-dhcp-server amd64 4.4.3
-P1-8 [1480 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 selinux-utils amd64 3.8.1-1
 [109 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 policycoreutils amd64 3.8.1
-2 [125 kB]
Fetched 1832 kB in 0s (6747 kB/s)
Preconfiguring packages ...
Selecting previously unselected package isc-dhcp-common.
(Reading database ... 10187 files and directories currently installed.)
Preparing to unpack .../isc-dhcp-common_4.4.3-P1-8_amd64.deb ...
Unpacking isc-dhcp-common (4.4.3-P1-8) ...
Selecting previously unselected package isc-dhcp-server.
Preparing to unpack .../isc-dhcp-server_4.4.3-P1-8_amd64.deb ...
Unpacking isc-dhcp-server (4.4.3-P1-8) ...
Selecting previously unselected package selinux-utils.
Preparing to unpack .../selinux-utils_3.8.1-1_amd64.deb ...
Unpacking selinux-utils (3.8.1-1) ...
Selecting previously unselected package policycoreutils.
Preparing to unpack .../policycoreutils_3.8.1-2_amd64.deb ...
Unpacking policycoreutils (3.8.1-2) ...
Setting up selinux-utils (3.8.1-1) ...
Setting up policycoreutils (3.8.1-2) ...
Setting up isc-dhcp-server (4.4.3-P1-8) ...
Generating /etc/default/isc-dhcp-server...
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of start.
Setting up isc-dhcp-common (4.4.3-P1-8) ...
===== Validasi konfigurasi =====
Internet Systems Consortium DHCP Server 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
Launching IPv4 server only.
Starting ISC DHCPv4 server: dhcpd.
===== Status service =====
Status of ISC DHCPv4 server: dhcpd is running.
root         491  0.0  0.0   9520  6920 ?        Ss   07:59   0:00 /usr/sbin/dhc
pd -4 -q -cf /etc/dhcp/dhcpd.conf eth0
root@tengen:~#
```

No 3

cat > /root/soal3.sh << 'EOF'

#!/bin/bash

Soal 3 - jichi: DHCP Relay ke tengen

echo "nameserver 192.168.122.1" > /etc/resolv.conf

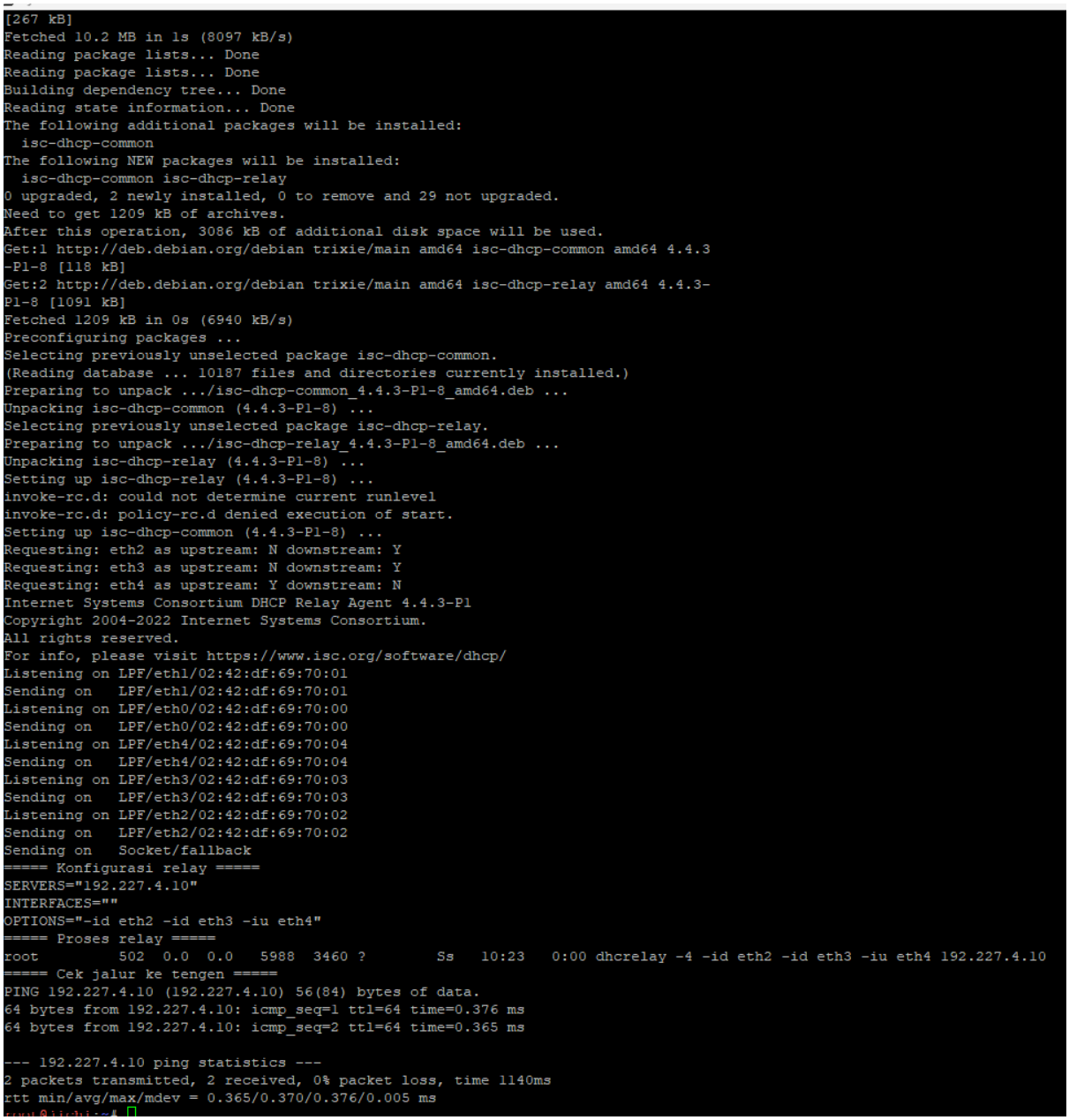
export DEBIAN_FRONTEND=noninteractive

command -v dhcrelay >/dev/null || (apt-get update && apt-get install -y isc-dhcp-relay)

```
cat > /etc/default/isc-dhcp-relay << EOT
SERVERS="192.227.4.10"
INTERFACES=""
OPTIONS="-id eth2 -id eth3 -iu eth4"
EOT

pkill dhcrelay 2>/dev/null; sleep 1
# -id = interface ke klien (Network 2 & 3), -iu = interface ke DHCP server (Network 4)
dhcrelay -4 -id eth2 -id eth3 -iu eth4 192.227.4.10

echo "===== Konfigurasi relay ====="
cat /etc/default/isc-dhcp-relay
echo "===== Proses relay ====="
ps aux | grep [d]hcrelay
echo "===== Cek jalur ke tengen ====="
ping -c 2 192.227.4.10
EOF
bash /root/soal1.sh ; bash /root/soal3.sh
cp /root/soal3.sh /etc/network/soal3.sh
```



sh /root/dhcp.sh

```
itadori:~# sh /root/dhcp.sh
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.227.2.10, server 192.227.4.10
udhcpc: lease of 192.227.2.10 obtained from 192.227.4.10, lease time 600
===== Hasil lease =====
eth0             UNKNOWN        192.227.2.10/24
default via 192.227.2.1 dev eth0 metric 83254
192.227.2.0/24 dev eth0 proto kernel scope link src 192.227.2.10
nameserver 192.168.122.1
PING 192.227.2.1 (192.227.2.1) 56(84) bytes of data.
64 bytes from 192.227.2.1: icmp_seq=1 ttl=64 time=0.641 ms
64 bytes from 192.227.2.1: icmp_seq=2 ttl=64 time=0.487 ms
64 bytes from 192.227.2.1: icmp_seq=3 ttl=64 time=0.461 ms

--- 192.227.2.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2320ms
rtt min/avg/max/mdev = 0.461/0.529/0.641/0.079 ms
itadori:~#
```

No 4

```
# Pool 1 - Tim Dev (itadori, fushiguro)
class "pool1" { match hardware; }
subclass "pool1" 1:02:00:00:00:a1:01;   # itadori
subclass "pool1" 1:02:00:00:00:a1:02;   # fushiguro

# Pool 2 - Tim Support (zenin, okkotsu)
class "pool2" { match hardware; }
subclass "pool2" 1:02:00:00:00:a2:01;   # zenin
subclass "pool2" 1:02:00:00:00:a2:02;   # okkotsu

# Network 4 - subnet lokal tengen, tanpa pool dinamis
subnet $P.4.0 netmask 255.255.255.0 {
}

# Network 2 - Divisi A (.36 - .69 sengaja tidak dialokasikan)
subnet $P.2.0 netmask 255.255.255.0 {
    option routers $P.2.1;
    option broadcast-address $P.2.255;
    option domain-name-servers 192.168.122.1;
    pool {
        range $P.2.10 $P.2.35;
        allow members of "pool1";
    }
    pool {
        range $P.2.70 $P.2.95;
        allow members of "pool2";
    }
}

# Network 3 - Divisi B
subnet $P.3.0 netmask 255.255.255.0 {
    range $P.3.10 $P.3.100;
    option routers $P.3.1;
    option broadcast-address $P.3.255;
    option domain-name-servers 192.168.122.1;
cp /root/soal4.sh /etc/network/soal4.shl; pkill dhcpd; rm -f /var/run/dhcpd.pid
===== Validasi konfigurasi =====
Internet Systems Consortium DHCP Server 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
Stopping ISC DHCPv4 server: dhcpd.
Launching IPv4 server only.
Starting ISC DHCPv4 server: dhcpd.
Status of ISC DHCPv4 server: dhcpd is running.
root@tengen:~#
```

```
# Minta IP lewat DHCP (melalui relay jichi), dengan MAC tetap untuk class DHCP
case "$(hostname)" in
  itadori)   MAC=02:00:00:00:a1:01 ;;
  fushiguro) MAC=02:00:00:00:a1:02 ;;
  zenin)     MAC=02:00:00:00:a2:01 ;;
  okkotsu)   MAC=02:00:00:00:a2:02 ;;
esac
ip addr flush dev eth0
if [ -n "$MAC" ]; then
  ip link set eth0 down
  ip link set eth0 address $MAC
fi
ip link set eth0 up
udhcpc -i eth0 -n -q
echo "===== Hasil lease ====="
ip link show eth0 | grep ether
ip -br addr show eth0
ip route
ping -c 3 $(ip route | awk '/default/ {print $3}')
EOF
sh /root/dhcp.sh
cp /root/dhcp.sh /etc/network/dhcp.sh
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.227.2.71, server 192.227.4.10
udhcpc: lease of 192.227.2.71 obtained from 192.227.4.10, lease time 600
===== Hasil lease =====
    link/ether 02:00:00:00:a2:02 brd ff:ff:ff:ff:ff:ff
eth0             UNKNOWN        192.227.2.71/24 fe80::ff:fe00:a202/64
default via 192.227.2.1 dev eth0 metric 83267
192.227.2.0/24 dev eth0 proto kernel scope link src 192.227.2.71
PING 192.227.2.1 (192.227.2.1) 56(84) bytes of data.
64 bytes from 192.227.2.1: icmp_seq=1 ttl=64 time=0.299 ms
64 bytes from 192.227.2.1: icmp_seq=2 ttl=64 time=0.414 ms
64 bytes from 192.227.2.1: icmp_seq=3 ttl=64 time=0.376 ms

--- 192.227.2.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2316ms
rtt min/avg/max/mdev = 0.299/0.363/0.414/0.047 ms
okkotsu:~#
```

cat > /root/soal4.sh << 'EOF'

#!/bin/bash

Soal 4 - tengen: 2 pool DHCP di Network 2 berdasarkan MAC (class)

P=192.227

echo "nameserver 192.168.122.1" > /etc/resolv.conf

command -v dhcpd >/dev/null || (apt-get update && apt-get install -y isc-dhcp-server)

echo 'INTERFACESv4="eth0"' > /etc/default/isc-dhcp-server

cat > /etc/dhcp/dhcpd.conf << EOT

authoritative;

```
default-lease-time 600;
max-lease-time 7200;

# ===== Class berdasarkan MAC address =====
# Pool 1 - Tim Dev (itadori, fushiguro)
class "pool1" { match hardware; }
subclass "pool1" 1:02:00:00:00:a1:01;   # itadori
subclass "pool1" 1:02:00:00:00:a1:02;   # fushiguro

# Pool 2 - Tim Support (zenin, okkotsu)
class "pool2" { match hardware; }
subclass "pool2" 1:02:00:00:00:a2:01;   # zenin
subclass "pool2" 1:02:00:00:00:a2:02;   # okkotsu

# Network 4 - subnet lokal tengen, tanpa pool dinamis
subnet $P.4.0 netmask 255.255.255.0 {
}

# Network 2 - Divisi A (.36 - .69 sengaja tidak dialokasikan)
subnet $P.2.0 netmask 255.255.255.0 {
   option routers $P.2.1;
   option broadcast-address $P.2.255;
   option domain-name-servers 192.168.122.1;
   pool {
      range $P.2.10 $P.2.35;
      allow members of "pool1";
   }
   pool {
      range $P.2.70 $P.2.95;
      allow members of "pool2";
   }
```

```
}

# Network 3 - Divisi B
subnet $P.3.0 netmask 255.255.255.0 {
   range $P.3.10 $P.3.100;
   option routers $P.3.1;
   option broadcast-address $P.3.255;
   option domain-name-servers 192.168.122.1;
}
EOT

echo "===== Validasi konfigurasi ====="
dhcpd -t -cf /etc/dhcp/dhcpd.conf || exit 1

# kosongkan lease lama supaya hasil uji bersih
> /var/lib/dhcp/dhcpd.leases
service isc-dhcp-server stop 2>/dev/null; pkill dhcpd; rm -f /var/run/dhcpd.pid
service isc-dhcp-server start
sleep 2
service isc-dhcp-server status
EOF
bash /root/soal4.sh
cp /root/soal4.sh /etc/network/soal4.sh



cat > /root/dhcp.sh << 'EOF'
#!/bin/sh
# Minta IP lewat DHCP (melalui relay jichi), dengan MAC tetap untuk class DHCP
case "$(hostname)" in
  itadori)   MAC=02:00:00:00:a1:01 ;;
```

```
  fushiguro) MAC=02:00:00:00:a1:02 ;;
  zenin)     MAC=02:00:00:00:a2:01 ;;
  okkotsu)   MAC=02:00:00:00:a2:02 ;;
esac
ip addr flush dev eth0
if [ -n "$MAC" ]; then
  ip link set eth0 down
  ip link set eth0 address $MAC
fi
ip link set eth0 up
udhcpc -i eth0 -n -q
echo "===== Hasil lease ====="
ip link show eth0 | grep ether
ip -br addr show eth0
ip route
ping -c 3 $(ip route | awk '/default/ {print $3}')
EOF
sh /root/dhcp.sh
cp /root/dhcp.sh /etc/network/dhcp.sh
```

no 5

```
lease 192.227.2.70 {
  starts 3 2026/10/14 10:44:55;
  ends 3 2026/10/14 10:54:55;
  hardware ethernet 02:00:00:00:a2:01;
lease 192.227.2.71 {
  starts 3 2026/10/14 10:45:08;
  ends 3 2026/10/14 10:55:08;
  hardware ethernet 02:00:00:00:a2:02;
lease 192.227.3.10 {
  starts 3 2026/10/14 10:50:38;
  ends 3 2026/10/14 11:20:38;
  hardware ethernet 02:00:00:00:b3:01;
lease 192.227.3.11 {
  starts 3 2026/10/14 10:51:01;
  ends 3 2026/10/14 11:21:01;
  hardware ethernet 02:00:00:00:b3:02;
lease 192.227.3.100 {
  starts 3 2026/10/14 10:51:10;
  ends 3 2026/10/14 22:51:10;
  hardware ethernet 02:00:00:00:b3:03;
root@tengen:~#
```

```
cat > /root/soal5.sh << 'EOF'
#!/bin/bash
# Soal 5 - tengen: short/long lease di Network 3 + DNS & domain-search
P=192.227
DOMAIN=k32.com
echo "nameserver 192.168.122.1" > /etc/resolv.conf
command -v dhcpd >/dev/null || (apt-get update && apt-get install -y isc-dhcp-server)
echo 'INTERFACESv4="eth0"' > /etc/default/isc-dhcp-server

cat > /etc/dhcp/dhcpd.conf << EOT
authoritative;
default-lease-time 600;
max-lease-time 7200;

# ===== Network 2: class berdasarkan MAC (soal 4) =====
class "pool1" { match hardware; }
subclass "pool1" 1:02:00:00:00:a1:01;   # itadori
```

```
subclass "pool1" 1:02:00:00:00:a1:02;   # fushiguro
class "pool2" { match hardware; }
subclass "pool2" 1:02:00:00:00:a2:01;   # zenin
subclass "pool2" 1:02:00:00:00:a2:02;   # okkotsu

# ===== Network 3: class berdasarkan MAC (soal 5) =====
class "short-lease" { match hardware; }
subclass "short-lease" 1:02:00:00:00:b3:01;   # ino
subclass "short-lease" 1:02:00:00:00:b3:02;   # kusakabe
class "long-lease" { match hardware; }
subclass "long-lease" 1:02:00:00:00:b3:03;    # miwa

# Network 4 - subnet lokal tengen, tanpa pool dinamis
subnet $P.4.0 netmask 255.255.255.0 {
}

# Network 2 - Divisi A (.36 - .69 tidak dialokasikan)
subnet $P.2.0 netmask 255.255.255.0 {
   option routers $P.2.1;
   option broadcast-address $P.2.255;
   option domain-name-servers 192.168.122.1;
   pool {
      range $P.2.10 $P.2.35;
      allow members of "pool1";
   }
   pool {
      range $P.2.70 $P.2.95;
      allow members of "pool2";
   }
}
```

```
# Network 3 - Divisi B
subnet $P.3.0 netmask 255.255.255.0 {
   option routers $P.3.1;
   option broadcast-address $P.3.255;
   option domain-name-servers $P.1.2, $P.1.3;
   option domain-search "$DOMAIN";
   # short lease 30 menit - ino, kusakabe
   pool {
      range $P.3.10 $P.3.89;
      allow members of "short-lease";
      default-lease-time 1800;
      max-lease-time 1800;
   }
   # long lease 12 jam - miwa
   pool {
      range $P.3.100 $P.3.199;
      allow members of "long-lease";
      default-lease-time 43200;
      max-lease-time 43200;
   }
}
EOT

echo "===== Validasi konfigurasi ====="
dhcpd -t -cf /etc/dhcp/dhcpd.conf || exit 1
service isc-dhcp-server stop 2>/dev/null; pkill dhcpd; rm -f /var/run/dhcpd.pid
service isc-dhcp-server start
sleep 2
service isc-dhcp-server status
EOF
bash /root/soal5.sh
```

```
cp /root/soal5.sh /etc/network/soal5.sh
```

```
cat > /root/dhcp.sh << 'EOF'
#!/bin/sh
# Minta IP lewat DHCP (melalui relay jichi), dengan MAC tetap untuk class DHCP
case "$(hostname)" in
  itadori)   MAC=02:00:00:00:a1:01 ;;
  fushiguro) MAC=02:00:00:00:a1:02 ;;
  zenin)     MAC=02:00:00:00:a2:01 ;;
  okkotsu)   MAC=02:00:00:00:a2:02 ;;
  ino)       MAC=02:00:00:00:b3:01 ;;
  kusakabe)  MAC=02:00:00:00:b3:02 ;;
  miwa)      MAC=02:00:00:00:b3:03 ;;
  kugisaki)  MAC=02:00:00:00:b3:04 ;;
esac
ip addr flush dev eth0
if [ -n "$MAC" ]; then
  ip link set eth0 down
  ip link set eth0 address $MAC
fi
ip link set eth0 up
udhcpc -i eth0 -n -q -O search -O domain
echo "===== Hasil lease ====="
ip link show eth0 | grep ether
ip -br addr show eth0
ip route
echo "----- /etc/resolv.conf -----"
cat /etc/resolv.conf
ping -c 3 $(ip route | awk '/default/ {print $3}')
```

```
EOF
sh /root/dhcp.sh
cp /root/dhcp.sh /etc/network/dhcp.sh
```

no 6

```
cat > /root/soal6.sh << 'EOF'
#!/bin/bash
# Soal 6 - tengen: static lease (fixed address) untuk kugisaki
P=192.227
bash /root/soal5.sh > /dev/null

cat >> /etc/dhcp/dhcpd.conf << EOT

# ===== Static lease - kugisaki (VIP) =====
# .3.99 berada di luar pool short (.10-.89) dan pool long (.100-.199)
host kugisaki {
   hardware ethernet 02:00:00:00:b3:04;
   fixed-address $P.3.99;
   option routers $P.3.1;
   option domain-name-servers $P.1.2, $P.1.3;
   option domain-search "k32.com";
}
EOT

echo "===== Validasi konfigurasi ====="
dhcpd -t -cf /etc/dhcp/dhcpd.conf || exit 1
service isc-dhcp-server stop 2>/dev/null; pkill dhcpd; rm -f /var/run/dhcpd.pid
service isc-dhcp-server start
sleep 2
```

service isc-dhcp-server status

echo "===== Reservasi kugisaki ====="

grep -A6 "host kugisaki" /etc/dhcp/dhcpd.conf

EOF

bash /root/soal6.sh

cp /root/soal6.sh /etc/network/soal6.sh

```
# .3.99 berada di luar pool short (.10-.89) dan pool long (.100-.199)
host kugisaki {
    hardware ethernet 02:00:00:00:b3:04;
    fixed-address $P.3.99;
    option routers $P.3.1;
    option domain-name-servers $P.1.2, $P.1.3;
    option domain-search "k32.com";
}
EOT

echo "===== Validasi konfigurasi ====="
dhcpd -t -cf /etc/dhcp/dhcpd.conf || exit 1
service isc-dhcp-server stop 2>/dev/null; pkill dhcpd; rm -f /var/run/dhcpd.pid
service isc-dhcp-server start
sleep 2
service isc-dhcp-server status
echo "===== Reservasi kugisaki ====="
grep -A6 "host kugisaki" /etc/dhcp/dhcpd.conf
EOF
bash /root/soal6.sh
cp /root/soal6.sh /etc/network/soal6.sh
Internet Systems Consortium DHCP Server 4.4.3-P1
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For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
===== Validasi konfigurasi =====
Internet Systems Consortium DHCP Server 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
Stopping ISC DHCPv4 server: dhcpd.
Launching IPv4 server only.
Starting ISC DHCPv4 server: dhcpd.
Status of ISC DHCPv4 server: dhcpd is running.
===== Reservasi kugisaki =====
host kugisaki {
    hardware ethernet 02:00:00:00:b3:04;
    fixed-address 192.227.3.99;
    option routers 192.227.3.1;
    option domain-name-servers 192.227.1.2, 192.227.1.3;
    option domain-search "k32.com";
}
```

```
kugisaki:~# sh /root/dhcp.sh
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.227.3.99, server 192.227.4.10
udhcpc: lease of 192.227.3.99 obtained from 192.227.4.10, lease time 600
===== Hasil lease =====
    link/ether 02:00:00:00:b3:04 brd ff:ff:ff:ff:ff:ff
eth0             UNKNOWN        192.227.3.99/24 fe80::ff:fe00:b304/64
default via 192.227.3.1 dev eth0 metric 83280
192.227.3.0/24 dev eth0 proto kernel scope link src 192.227.3.99
----- /etc/resolv.conf -----
search k32.com
nameserver 192.227.1.2
nameserver 192.227.1.3
PING 192.227.3.1 (192.227.3.1) 56(84) bytes of data.
64 bytes from 192.227.3.1: icmp_seq=1 ttl=64 time=0.606 ms
64 bytes from 192.227.3.1: icmp_seq=2 ttl=64 time=1.29 ms
64 bytes from 192.227.3.1: icmp_seq=3 ttl=64 time=0.494 ms

--- 192.227.3.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2246ms
rtt min/avg/max/mdev = 0.494/0.796/1.290/0.351 ms
kugisaki:~#
```

sh /root/dhcp.sh

```
rtt min/avg/max/mdev = 0.494/0.796/1.290/0.351 ms
kugisaki:~# echo "===== Renew lease ====="
udhcpc -i eth0 -n -q -O search -O domain
ip -br addr show eth0
===== Renew lease =====
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.227.3.99, server 192.227.4.10
udhcpc: lease of 192.227.3.99 obtained from 192.227.4.10, lease time 600
eth0             UNKNOWN        192.227.3.99/24 fe80::ff:fe00:b304/64
kugisaki:~#
```

No 7

cat > /root/soal7.sh << 'EOF'

#!/bin/bash

Soal 7 - tengen: samakan DNS & domain-search untuk Network 2

bash /root/soal6.sh > /dev/null

sed -i 's|option domain-name-servers 192.168.122.1;|option domain-name-servers 192.227.1.2, 192.227.1.3;\n   option domain-search "k32.com";|' /etc/dhcp/dhcpd.conf

echo "===== Validasi konfigurasi ====="

dhcpd -t -cf /etc/dhcp/dhcpd.conf || exit 1

```
service isc-dhcp-server stop 2>/dev/null; pkill dhcpd; rm -f /var/run/dhcpd.pid
service isc-dhcp-server start
sleep 2
service isc-dhcp-server status
echo "===== Opsi per subnet ====="
grep -E "subnet|routers|broadcast|domain-name-servers|domain-search" /etc/dhcp/dhcpd.conf
EOF

cat > /root/cek7.sh << 'EOF'
#!/bin/bash
# Soal 7 - tengen: cocokkan lease aktif dengan kebijakan pool & cek bentrok alamat
LEASES=${1:-/var/lib/dhcp/dhcpd.leases}
awk '
 /^lease /           { ip=$2; mac="-"; st="-" }
 $1=="binding"       { st=$3; sub(";","",st) }
 /hardware ethernet/  { mac=$3; sub(";","",mac) }
 /^}/                { if (ip!="") L[ip]=mac" "st; ip="" }
 END { for (i in L) print i, L[i] }' "$LEASES" | sort -t. -k3,3n -k4,4n > /tmp/lease_now

printf "%-16s %-19s %-10s %-9s %-14s %s\n" IP MAC KLIEN STATUS POOL HASIL
ERR=0
while read -r ip mac st; do
 [ "$st" = "active" ] || continue
 net=$(echo "$ip" | cut -d. -f3); h=$(echo "$ip" | cut -d. -f4)
 case "$mac" in
   02:00:00:00:a1:01) n=itadori;   lo=10;  hi=35;  want=2 ;;
   02:00:00:00:a1:02) n=fushiguro; lo=10;  hi=35;  want=2 ;;
   02:00:00:00:a2:01) n=zenin;     lo=70;  hi=95;  want=2 ;;
   02:00:00:00:a2:02) n=okkotsu;   lo=70;  hi=95;  want=2 ;;
   02:00:00:00:b3:01) n=ino;       lo=10;  hi=89;  want=3 ;;
   02:00:00:00:b3:02) n=kusakabe;  lo=10;  hi=89;  want=3 ;;
```

```
   02:00:00:00:b3:03) n=miwa;      lo=100; hi=199; want=3 ;;
   *)                 n="?";       lo=0;   hi=-1;  want=0 ;;
  esac
  if [ "$net" = "$want" ] && [ "$h" -ge "$lo" ] && [ "$h" -le "$hi" ]; then
    res="OK"
  else
    res="DI LUAR POOL"; ERR=1
  fi
  printf "%-16s %-19s %-10s %-9s %-14s %s\n" "$ip" "$mac" "$n" "$st" ".$want.$lo-.$want.$hi" "$res"
done < /tmp/lease_now

echo
DUP_IP=$(awk '$3=="active"{print $1}' /tmp/lease_now | sort | uniq -d)
DUP_MAC=$(awk '$3=="active"{print $2}' /tmp/lease_now | sort | uniq -d)
if [ -z "$DUP_IP$DUP_MAC" ]; then
  echo "Bentrok alamat : TIDAK ADA (setiap IP & MAC aktif unik)"
else
  echo "Bentrok alamat : ADA -> IP: $DUP_IP  MAC: $DUP_MAC"; ERR=1
fi
if awk '$3=="active"{print $1}' /tmp/lease_now | grep -q '\.3\.99$'; then
  echo "Fixed address  : .3.99 ikut terpakai sebagai lease dinamis (SALAH)"; ERR=1
else
  echo "Fixed address  : .3.99 tidak dipakai pool dinamis (khusus kugisaki)"
fi
[ $ERR -eq 0 ] && echo "KESIMPULAN     : semua lease sesuai kebijakan pool" \
               || echo "KESIMPULAN     : ada lease yang melanggar kebijakan"
EOF
bash /root/soal7.sh
cp /root/soal7.sh /root/cek7.sh /etc/network/
```

```
    res="DI LUAR POOL"; ERR=1
  fi
  printf "%-16s %-19s %-10s %-9s %-14s %s\n" "$ip" "$mac" "$n" "$st" ".$want.$lo-.$want.$hi" "$res"
done < /tmp/lease_now

cp /root/soal7.sh /root/cek7.sh /etc/network/lease yang melanggar kebijakan"\ERR=1
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For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
Internet Systems Consortium DHCP Server 4.4.3-P1
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For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
===== Validasi konfigurasi =====
Internet Systems Consortium DHCP Server 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
Stopping ISC DHCPv4 server: dhcpd.
Launching IPv4 server only.
Starting ISC DHCPv4 server: dhcpd.
Status of ISC DHCPv4 server: dhcpd is running.
===== Opsi per subnet =====
# Network 4 - subnet lokal tengen, tanpa pool dinamis
subnet 192.227.4.0 netmask 255.255.255.0 {
subnet 192.227.2.0 netmask 255.255.255.0 {
    option routers 192.227.2.1;
    option broadcast-address 192.227.2.255;
    option domain-name-servers 192.227.1.2, 192.227.1.3;
    option domain-search "k32.com";
subnet 192.227.3.0 netmask 255.255.255.0 {
    option routers 192.227.3.1;
    option broadcast-address 192.227.3.255;
    option domain-name-servers 192.227.1.2, 192.227.1.3;
    option domain-search "k32.com";
    option routers 192.227.3.1;
    option domain-name-servers 192.227.1.2, 192.227.1.3;
    option domain-search "k32.com";
```

```
===== itadori =====
    link/ether 02:00:00:00:a1:01 brd ff:ff:ff:ff:ff:ff
    inet 192.227.2.10/24 brd 192.227.2.255 scope global eth0
default via 192.227.2.1 dev eth0 metric 83254
192.227.2.0/24 dev eth0 proto kernel scope link src 192.227.2.10
nameserver 192.227.1.2
nameserver 192.227.1.3
itadori:~#
```

```
4 bytes from 192.227.2.1: icmp_seq=3 ttl=64 time=0.376 ms

-- 192.227.2.1 ping statistics ---
 packets transmitted, 3 received, 0% packet loss, time 2316ms
tt min/avg/max/mdev = 0.299/0.363/0.414/0.047 ms
kkotsu:~# sed -i 's/udhcpc -i eth0 -n -q$/udhcpc -i eth0 -n -q -O search -O dom
in/' /root/dhcp.sh
p /root/dhcp.sh /etc/network/dhcp.sh
h /root/dhcp.sh > /dev/null 2>&1
lear
cho "===== $(hostname) ====="
p addr show eth0 | grep -E "ether|inet "
p route
at /etc/resolv.conf
==== okkotsu =====
   link/ether 02:00:00:00:a2:02 brd ff:ff:ff:ff:ff:ff
   inet 192.227.2.71/24 brd 192.227.2.255 scope global eth0
efault via 192.227.2.1 dev eth0 metric 83267
92.227.2.0/24 dev eth0 proto kernel scope link src 192.227.2.71
earch k32.com
ameserver 192.227.1.2
ameserver 192.227.1.3
kkotsu:~#
```

```
root@tengen:~# bash -n /root/cek7.sh && echo "cek7.sh OK"
ls /etc/network
cek7.sh OK
cek7.sh  if-down.d  if-post-down.d  if-pre-up.d  if-up.d  interfaces  interfaces.d  soal1.sh  soal2.sh  soal4.sh  soal5.sh  soal6.sh  soal7.sh
root@tengen:~#
```

```
IP              MAC                KLIEN      STATUS    POOL           HASIL
192.227.2.10    02:00:00:00:a1:01  itadori    active    .2.10-.2.35    OK
192.227.2.11    02:00:00:00:a1:02  fushiguro  active    .2.10-.2.35    OK
192.227.2.70    02:00:00:00:a2:01  zenin      active    .2.70-.2.95    OK
192.227.2.71    02:00:00:00:a2:02  okkotsu    active    .2.70-.2.95    OK
192.227.3.100   02:00:00:00:b3:03  miwa       active    .3.100-.3.199  OK

Bentrok alamat : TIDAK ADA (setiap IP & MAC aktif unik)
Fixed address  : .3.99 tidak dipakai pool dinamis (khusus kugisaki)
KESIMPULAN     : semua lease sesuai kebijakan pool
root@tengen:~#
```

No 8

```
root@tengen:~# cat > /root/soal8.sh << 'EOF'
#!/bin/bash
# Soal 8 - tengen: capture DHCP DORA (lewat relay jichi)
command -v tcpdump >/dev/null || (apt-get update && apt-get install -y tcpdump)
echo ">>> Menunggu 4 paket DHCP... sekarang jalankan 'sh /root/dhcp.sh' di client"
tcpdump -i eth0 -n -vv -c 4 -l 'udp port 67 or udp port 68' 2>&1 | tee /root/soal8_capture.txt
echo
echo "================ RINGKASAN ================"
grep -E "IP [0-9]|DHCP-Message|Gateway-IP|Your-IP|Server-ID|Lease-Time|Client-Ethernet"
/root/soal8_capture.txt
EOF
cp /root/soal8.sh /etc/network/soal8.sh
bash /root/soal8.sh
>>> Menunggu 4 paket DHCP... sekarang jalankan 'sh /root/dhcp.sh' di client
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
22:35:21.666226 IP (tos 0x0, ttl 64, id 63358, offset 0, flags [DF], proto UDP (17), length 328)
    192.227.4.1.67 > 192.227.4.10.67: [udp sum ok] BOOTP/DHCP, Request from 02:00:00:00:a1:01,
length 300, hops 1, xid 0x981a035b, Flags [none] (0x0000)
          Gateway-IP 192.227.2.1
          Client-Ethernet-Address 02:00:00:00:a1:01
          Vendor-rfc1048 Extensions
            Magic Cookie 0x63825363
            DHCP-Message (53), length 1: Discover
            MSZ (57), length 2: 576
            Parameter-Request (55), length 8:
              Subnet-Mask (1), Default-Gateway (3), Domain-Name-Server (6), Hostname (12)
              Domain-Name (15), BR (28), NTP (42), Unknown (119)
          Vendor-Class (60), length 12: "udhcp 1.37.0"
          Client-ID (61), length 7: ether 02:00:00:00:a1:01
22:35:21.666632 IP (tos 0x0, ttl 64, id 37379, offset 0, flags [DF], proto UDP (17), length 328)
```

192.227.4.10.67 > 192.227.2.1.67: [udp sum ok] BOOTP/DHCP, Reply, length 300, hops 1, xid 0x981a035b, Flags [none] (0x0000)

Your-IP 192.227.2.10

Gateway-IP 192.227.2.1

Client-Ethernet-Address 02:00:00:00:a1:01

Vendor-rfc1048 Extensions

Magic Cookie 0x63825363

DHCP-Message (53), length 1: Offer

Server-ID (54), length 4: 192.227.4.10

Lease-Time (51), length 4: 600

Subnet-Mask (1), length 4: 255.255.255.0

Default-Gateway (3), length 4: 192.227.2.1

Domain-Name-Server (6), length 8: 192.227.1.2,192.227.1.3

BR (28), length 4: 192.227.2.255

Unknown (119), length 9: 3.107.51.50.3.99.111.109.0

22:35:21.679161 IP (tos 0x0, ttl 64, id 63359, offset 0, flags [DF], proto UDP (17), length 328)

192.227.4.1.67 > 192.227.4.10.67: [udp sum ok] BOOTP/DHCP, Request from 02:00:00:00:a1:01, length 300, hops 1, xid 0x981a035b, Flags [none] (0x0000)

Gateway-IP 192.227.2.1

Client-Ethernet-Address 02:00:00:00:a1:01

Vendor-rfc1048 Extensions

Magic Cookie 0x63825363

DHCP-Message (53), length 1: Request

Requested-IP (50), length 4: 192.227.2.10

Server-ID (54), length 4: 192.227.4.10

MSZ (57), length 2: 576

Parameter-Request (55), length 8:

Subnet-Mask (1), Default-Gateway (3), Domain-Name-Server (6), Hostname (12)

Domain-Name (15), BR (28), NTP (42), Unknown (119)

Vendor-Class (60), length 12: "udhcp 1.37.0"

Client-ID (61), length 7: ether 02:00:00:00:a1:01

22:35:21.680220 IP (tos 0x0, ttl 64, id 37382, offset 0, flags [DF], proto UDP (17), length 328)

192.227.4.10.67 > 192.227.2.1.67: [udp sum ok] BOOTP/DHCP, Reply, length 300, hops 1, xid 0x981a035b, Flags [none] (0x0000)

Your-IP 192.227.2.10

Gateway-IP 192.227.2.1

Client-Ethernet-Address 02:00:00:00:a1:01

Vendor-rfc1048 Extensions

Magic Cookie 0x63825363

DHCP-Message (53), length 1: ACK

Server-ID (54), length 4: 192.227.4.10

Lease-Time (51), length 4: 600

Subnet-Mask (1), length 4: 255.255.255.0

Default-Gateway (3), length 4: 192.227.2.1

Domain-Name-Server (6), length 8: 192.227.1.2,192.227.1.3

BR (28), length 4: 192.227.2.255

Unknown (119), length 9: 3.107.51.50.3.99.111.109.0

4 packets captured

4 packets received by filter

0 packets dropped by kernel

========== RINGKASAN ==========

Gateway-IP 192.227.2.1

Client-Ethernet-Address 02:00:00:00:a1:01

DHCP-Message (53), length 1: Discover

Your-IP 192.227.2.10

Gateway-IP 192.227.2.1

Client-Ethernet-Address 02:00:00:00:a1:01

DHCP-Message (53), length 1: Offer

Server-ID (54), length 4: 192.227.4.10

Lease-Time (51), length 4: 600

Gateway-IP 192.227.2.1

Client-Ethernet-Address 02:00:00:00:a1:01

DHCP-Message (53), length 1: Request

Server-ID (54), length 4: 192.227.4.10

Your-IP 192.227.2.10

Gateway-IP 192.227.2.1

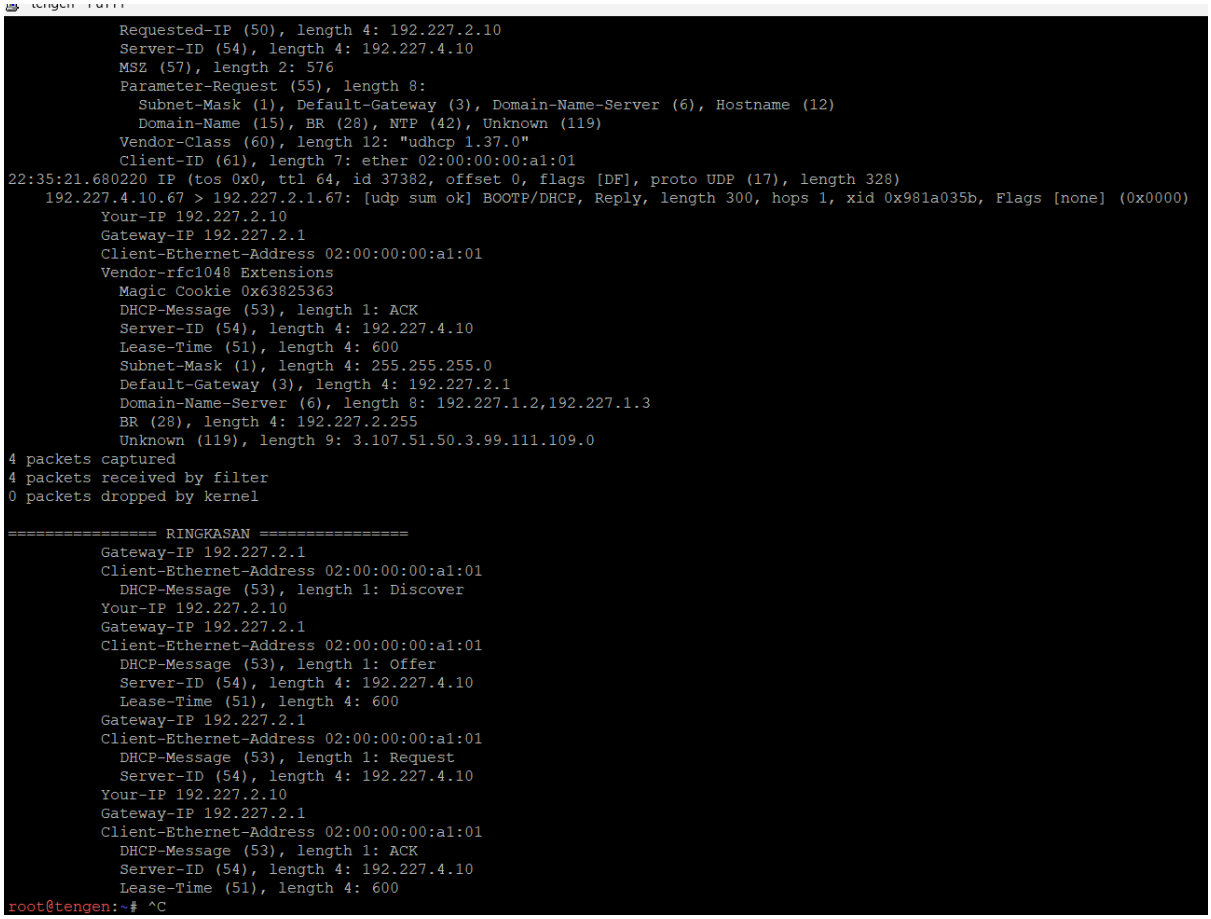
Client-Ethernet-Address 02:00:00:00:a1:01

DHCP-Message (53), length 1: ACK

Server-ID (54), length 4: 192.227.4.10

Lease-Time (51), length 4: 600

root@tengen:~#



```
root@jichi:~# cat > /root/soal9.sh << 'EOF'
#!/bin/bash
# Soal 9 - jichi: simulasi gangguan DHCP Relay
case "$1" in
  stop)
    service isc-dhcp-relay stop 2>/dev/null
    pkill dhcrelay
    sleep 1
    echo "===== Status relay setelah dihentikan ====="
    ps aux | grep [d]hcrelay || echo "dhcrelay TIDAK berjalan (relay mati)"
    ;;
  start)
    bash /root/soal3.sh
    ;;
  *)
    echo "Pakai: bash /root/soal9.sh stop | start"
    ;;
esac
EOF
cp /root/soal9.sh /etc/network/soal9.sh
bash /root/soal9.sh stop
===== Status relay setelah dihentikan =====
dhcrelay TIDAK berjalan (relay mati)
root@jichi:~#
```

No 9

itadori:~# sh /root/dhcp.sh

udhcpc: started, v1.37.0

udhcpc: broadcasting discover

udhcpc: broadcasting select for 192.227.2.10, server 192.227.4.10

udhcpc: lease of 192.227.2.10 obtained from 192.227.4.10, lease time 600

===== Hasil lease =====

link/ether 02:00:00:00:a1:01 brd ff:ff:ff:ff:ff:ff

eth0 UNKNOWN 192.227.2.10/24 fe80::ff:fe00:a101/64

default via 192.227.2.1 dev eth0 metric 83254

192.227.2.0/24 dev eth0 proto kernel scope link src 192.227.2.10

PING 192.227.2.1 (192.227.2.1) 56(84) bytes of data.

64 bytes from 192.227.2.1: icmp_seq=1 ttl=64 time=0.381 ms

64 bytes from 192.227.2.1: icmp_seq=2 ttl=64 time=0.403 ms

64 bytes from 192.227.2.1: icmp_seq=3 ttl=64 time=0.484 ms

--- 192.227.2.1 ping statistics ---

3 packets transmitted, 3 received, 0% packet loss, time 2284ms

```
rtt min/avg/max/mdev = 0.381/0.422/0.484/0.044 ms
itadori:~#


root@jichi:~#    bash /root/soal9.sh start
Requesting: eth2 as upstream: N downstream: Y
Requesting: eth3 as upstream: N downstream: Y
Requesting: eth4 as upstream: Y downstream: N
Internet Systems Consortium DHCP Relay Agent 4.4.3-P1
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All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Listening on LPF/eth1/02:42:df:69:70:01
Sending on   LPF/eth1/02:42:df:69:70:01
Listening on LPF/eth0/02:42:df:69:70:00
Sending on   LPF/eth0/02:42:df:69:70:00
Listening on LPF/eth4/02:42:df:69:70:04
Sending on   LPF/eth4/02:42:df:69:70:04
Listening on LPF/eth3/02:42:df:69:70:03
Sending on   LPF/eth3/02:42:df:69:70:03
Listening on LPF/eth2/02:42:df:69:70:02
Sending on   LPF/eth2/02:42:df:69:70:02
Sending on   Socket/fallback
===== Konfigurasi relay =====
SERVERS="192.227.4.10"
INTERFACES=""
OPTIONS="-id eth2 -id eth3 -iu eth4"
===== Proses relay =====
root       532  0.0  0.0   5988  3456 ?        Ss   22:54   0:00 dhcrelay -4 -id eth2 -id eth3 -iu eth4
192.227.4.10
===== Cek jalur ke tengen =====
PING 192.227.4.10 (192.227.4.10) 56(84) bytes of data.
```

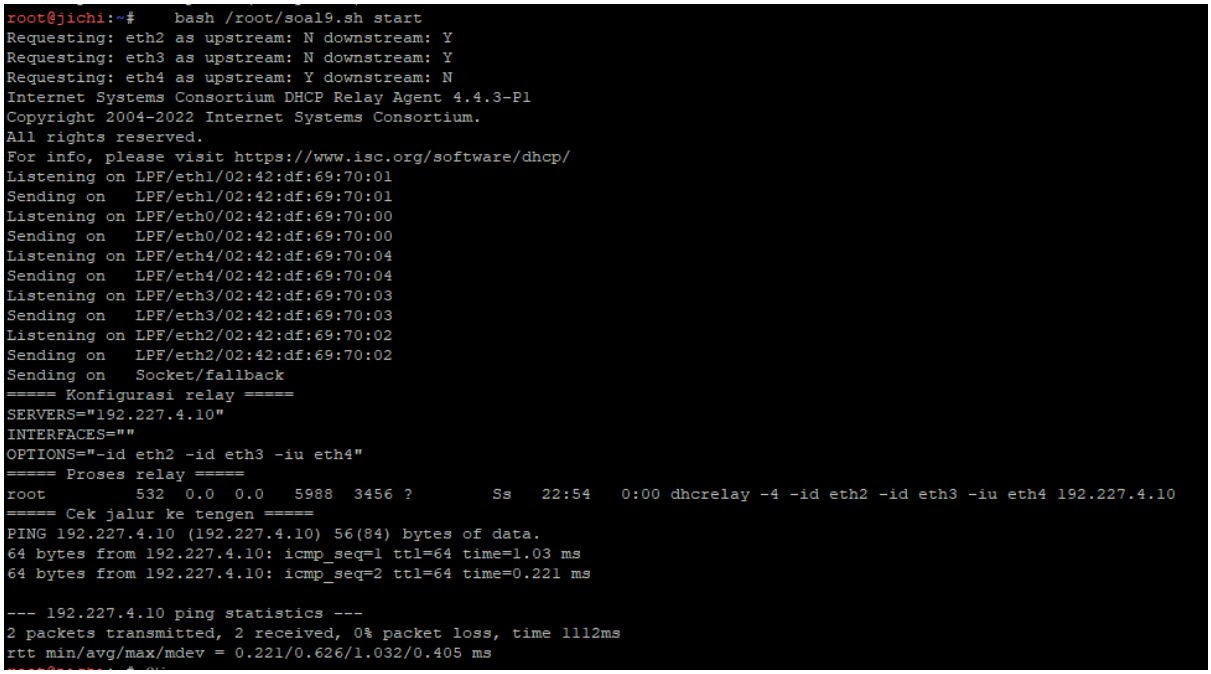
64 bytes from 192.227.4.10: icmp_seq=1 ttl=64 time=1.03 ms

64 bytes from 192.227.4.10: icmp_seq=2 ttl=64 time=0.221 ms

--- 192.227.4.10 ping statistics ---

2 packets transmitted, 2 received, 0% packet loss, time 1112ms

rtt min/avg/max/mdev = 0.221/0.626/1.032/0.405 ms



```
udhcpc: lease of 192.227.2.10 obtained from 192.227.4.10, lease time 600
===== Hasil lease =====
    link/ether 02:00:00:00:a1:01 brd ff:ff:ff:ff:ff:ff
eth0             UNKNOWN        192.227.2.10/24 fe80::ff:fe00:a101/64
default via 192.227.2.1 dev eth0 metric 83254
192.227.2.0/24 dev eth0 proto kernel scope link src 192.227.2.10
PING 192.227.2.1 (192.227.2.1) 56(84) bytes of data.
64 bytes from 192.227.2.1: icmp_seq=1 ttl=64 time=0.431 ms
64 bytes from 192.227.2.1: icmp_seq=2 ttl=64 time=0.554 ms
64 bytes from 192.227.2.1: icmp_seq=3 ttl=64 time=0.463 ms

--- 192.227.2.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2257ms
rtt min/avg/max/mdev = 0.431/0.482/0.554/0.052 ms
itadori:~# sh /root/dhcp.sh
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc failed to get a DHCP lease
udhcpc: no lease, failing
===== Hasil lease =====
    link/ether 02:00:00:00:a1:01 brd ff:ff:ff:ff:ff:ff
eth0             UNKNOWN        fe80::ff:fe00:a101/64
ping: usage error: Destination address required
itadori:~# ^C
itadori:~#    sh /root/dhcp.sh
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.227.2.10, server 192.227.4.10
udhcpc: lease of 192.227.2.10 obtained from 192.227.4.10, lease time 600
===== Hasil lease =====
    link/ether 02:00:00:00:a1:01 brd ff:ff:ff:ff:ff:ff
eth0             UNKNOWN        192.227.2.10/24 fe80::ff:fe00:a101/64
default via 192.227.2.1 dev eth0 metric 83254
192.227.2.0/24 dev eth0 proto kernel scope link src 192.227.2.10
PING 192.227.2.1 (192.227.2.1) 56(84) bytes of data.
64 bytes from 192.227.2.1: icmp_seq=1 ttl=64 time=0.381 ms
64 bytes from 192.227.2.1: icmp_seq=2 ttl=64 time=0.403 ms
64 bytes from 192.227.2.1: icmp_seq=3 ttl=64 time=0.484 ms

--- 192.227.2.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2284ms
rtt min/avg/max/mdev = 0.381/0.422/0.484/0.044 ms
```

No 10

```
root@jichi:~# cat > /root/soal10.sh << 'EOF'
#!/bin/bash
# Soal 10 - autostart DHCP Relay (jichi) & DHCP Server (tengen) setelah restart

# 1) Script yang dijalankan otomatis setiap node boot
cat > /etc/network/autostart.sh << 'EOT'
#!/bin/bash
echo "===== autostart $(hostname) $(date) ====="
sleep 3
cp -n /etc/network/*.sh /root/ 2>/dev/null
case "$(hostname)" in
  jichi)
    bash /root/soal1.sh
    bash /root/soal3.sh
    ;;
  tengen)
    bash /root/soal1.sh
    # tunggu jichi siap (internet dibutuhkan bila paket harus diinstal ulang)
p 2; done in $(seq 1 60); do ping -c1 -W1 8.8.8.8 >/dev/null 2>&1 && break; sleep
    bash /root/soal7.sh
    ;;
esac
echo "===== autostart selesai $(date) ====="
bash /root/soal10.shostart.shstart.sh ====="/dev/null/null ;;e saat boot/dev/nul
===== /etc/network/interfaces =====
#
# This is a sample network config, please uncomment lines to configure the network
#

# Uncomment this line to load custom interface files
```

```
# source /etc/network/interfaces.d/*

# Static config for eth0
#auto eth0
#iface eth0 inet static
#       address 192.168.0.2
#       netmask 255.255.255.0
#       gateway 192.168.0.1
#       up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#       hostname debinet-1

# Static config for eth1
#auto eth1
#iface eth1 inet static
#       address 192.168.1.2
#       netmask 255.255.255.0
#       gateway 192.168.1.1
#       up echo nameserver 192.168.1.1 > /etc/resolv.conf

# DHCP config for eth1
#auto eth1
#iface eth1 inet dhcp
#       hostname debinet-1

# Static config for eth2
#auto eth2
#iface eth2 inet static
```

```
#       address 192.168.2.2
#       netmask 255.255.255.0
#       gateway 192.168.2.1
#       up echo nameserver 192.168.2.1 > /etc/resolv.conf

# DHCP config for eth2
#auto eth2
#iface eth2 inet dhcp
#       hostname debinet-1

# Static config for eth3
#auto eth3
#iface eth3 inet static
#       address 192.168.3.2
#       netmask 255.255.255.0
#       gateway 192.168.3.1
#       up echo nameserver 192.168.3.1 > /etc/resolv.conf

# DHCP config for eth3
#auto eth3
#iface eth3 inet dhcp
#       hostname debinet-1

# Soal 10 - autostart layanan DHCP saat node boot
auto lo
iface lo inet loopback
    up setsid /etc/network/autostart.sh > /var/log/autostart.log 2>&1 < /dev/null &
===== /etc/network/autostart.sh =====
#!/bin/bash
echo "===== autostart $(hostname) $(date) ====="
sleep 3
```

```
cp -n /etc/network/*.sh /root/ 2>/dev/null
case "$(hostname)" in
  jichi)
    bash /root/soal1.sh
    bash /root/soal3.sh
    ;;
  tengen)
    bash /root/soal1.sh
    # tunggu jichi siap (internet dibutuhkan bila paket harus diinstal ulang)
    for i in $(seq 1 60); do ping -c1 -W1 8.8.8.8 >/dev/null 2>&1 && break; sleep 2; done
    bash /root/soal7.sh
    ;;
esac
echo "===== autostart selesai $(date) ====="
root@jichi:~# ls /etc/network
autostart.sh  if-post-down.d  if-up.d     interfaces.d  soal10.sh  soal9.sh
if-down.d     if-pre-up.d     interfaces  soal1.sh      soal3.sh
root@jichi:~#
```

```
root@tengen:~# ls /etc/network
cek7.sh  if-down.d  if-post-down.d  if-pre-up.d  if-up.d  interfaces  interfaces.d  soal1.sh  soal2.sh
soal4.sh  soal5.sh  soal6.sh  soal7.sh  soal8.sh
root@tengen:~#
```

root@tengen:~# ls /etc/network ; tail -4 /etc/network/interfaces

cek7.sh if-down.d if-post-down.d if-pre-up.d if-up.d interfaces interfaces.d soal1.sh soal2.sh soal4.sh soal5.sh soal6.sh soal7.sh soal8.sh

DHCP config for eth3

#auto eth3

#iface eth3 inet dhcp

hostname debinet-1

root@tengen:~#

Konfigurasi

/etc/network/autostart.sh (tengen):

bash

#!/bin/bash

echo "===== autostart $(hostname) $(date) ====="

sleep 3

cp -n /etc/network/*.sh /root/ 2>/dev/null

bash /root/soal1.sh

for i in $(seq 1 60); do ping -c1 -W1 8.8.8.8 >/dev/null 2>&1 && break; sleep 2; done

bash /root/soal7.sh

echo "===== autostart selesai $(date) ====="

Di jichi, isinya menjalankan soal1.sh lalu soal3.sh.

Hook boot di /etc/network/interfaces (jichi dan tengen):

root@jichi:~# sed -i 's/^iface lo inet loopback$/iface lo inet manual/' /etc/network/interfaces

root@jichi:~# tail -4 /etc/network/interfaces

Soal 10 - autostart layanan DHCP saat node boot

auto lo

iface lo inet manual

  up setsid /etc/network/autostart.sh > /var/log/autostart.log 2>&1 < /dev/null &

Ditambah update-rc.d isc-dhcp-relay defaults di jichi dan update-rc.d isc-dhcp-server defaults di tengen.

Sebelum diperbaiki (restart dengan inet loopback)

root@jichi:~# ps -o lstart= -p 1; ip -br addr; ip route | grep default; ps aux | grep [d]hcrelay; tail -5 /var/log/autostart.log

Wed Oct 14 23:11:21 2026

lo UNKNOWN 127.0.0.1/8 ::1/128

eth0 UNKNOWN fe80::42:dfff:fe69:7000/64

eth1 UNKNOWN fe80::42:dfff:fe69:7001/64

eth2 UNKNOWN fe80::42:dfff:fe69:7002/64

eth3 UNKNOWN fe80::42:dfff:fe69:7003/64

eth4 UNKNOWN fe80::42:dfff:fe69:7004/64

tail: cannot open '/var/log/autostart.log' for reading: No such file or directory

Setelah diperbaiki (restart dengan inet manual)

jichi:

Wed Oct 14 23:13:00 2026

eth0 UNKNOWN 192.168.122.121/24

eth1 UNKNOWN 192.227.1.1/24 fe80::42:dfff:fe69:7001/64

lo UNKNOWN 127.0.0.1/8 ::1/128

eth2 UNKNOWN 192.227.2.1/24 fe80::42:dfff:fe69:7002/64

eth3 UNKNOWN 192.227.3.1/24 fe80::42:dfff:fe69:7003/64

eth4 UNKNOWN 192.227.4.1/24 fe80::42:dfff:fe69:7004/64

default via 192.168.122.1 dev eth0 metric 87240

root 130 0.0 0.0 5988 3328 ? Ss 23:13 0:00 dhcrelay -4 -id eth2 -id eth3 -iu eth4 192.227.4.10

tengen:

Wed Oct 14 23:13:12 2026

eth0 UNKNOWN 192.227.4.10/24 fe80::42:94ff:fe19:5e00/64

default via 192.227.4.1 dev eth0

Status of ISC DHCPv4 server: dhcpd is running.

...

===== autostart selesai Wed Oct 14 23:13:35 UTC 2026 =====

Relay mendapat **PID baru 130 dengan waktu start 23:13**, sama dengan waktu boot container. Ini membuktikan relay dinyalakan oleh autostart, bukan dijalankan manual (sebelum restart, PID-nya 532).

```
IP              MAC                KLIEN      STATUS    POOL           HASIL

Bentrok alamat : TIDAK ADA (setiap IP & MAC aktif unik)
Fixed address  : .3.99 tidak dipakai pool dinamis (khusus kugisaki)
KESIMPULAN     : semua lease sesuai kebijakan pool
root@tengen:~# ^C
root@tengen:~#
```

```
IP              MAC                KLIEN      STATUS    POOL           HASIL
192.227.2.10    02:00:00:00:a1:01  itadori    active    .2.10-.2.35    OK
192.227.2.70    02:00:00:00:a2:01  zenin      active    .2.70-.2.95    OK
192.227.3.11    02:00:00:00:b3:02  kusakabe   active    .3.10-.3.89    OK
192.227.3.100   02:00:00:00:b3:03  miwa       active    .3.100-.3.199  OK

Bentrok alamat : TIDAK ADA (setiap IP & MAC aktif unik)
Fixed address  : .3.99 tidak dipakai pool dinamis (khusus kugisaki)
KESIMPULAN     : semua lease sesuai kebijakan pool
root@tengen:~#
```